

3)Cold Migration of VM from one host to another

Aim:

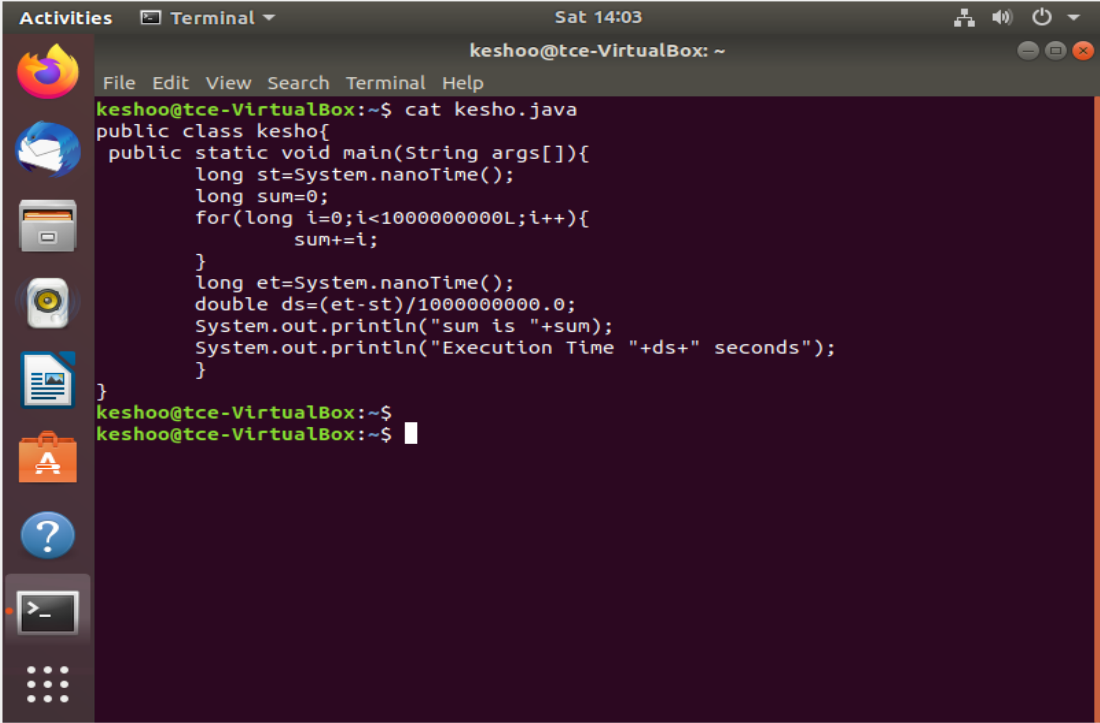
To perform cold migration of a virtual machine (VM) from one host to another using the export/import appliance method (OVA format) in order to transfer VM configurations and disk data seamlessly while the VM is powered off.

STEPS:

Prerequisite

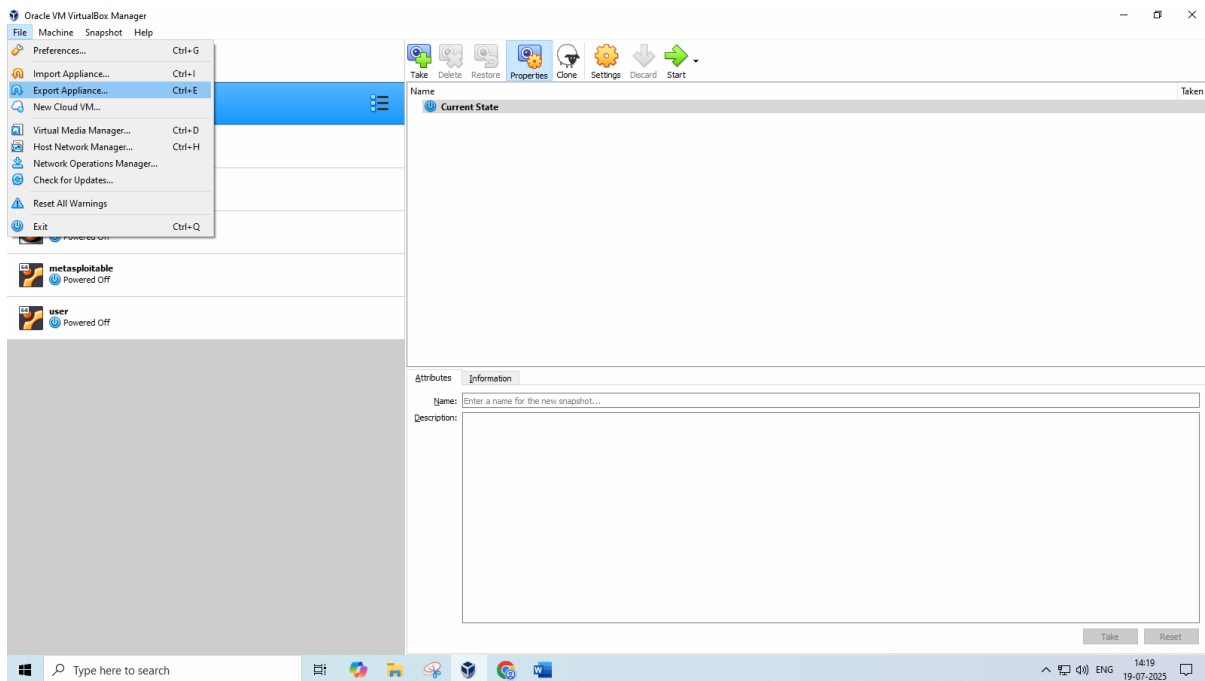
- VM must be shut down (powered off).
- Both systems (source and destination) must have VirtualBox
- Ensure enough storage on destination host.

1)Created a java file to check whether the same file exist in vm after imported(cold migrated) in another system

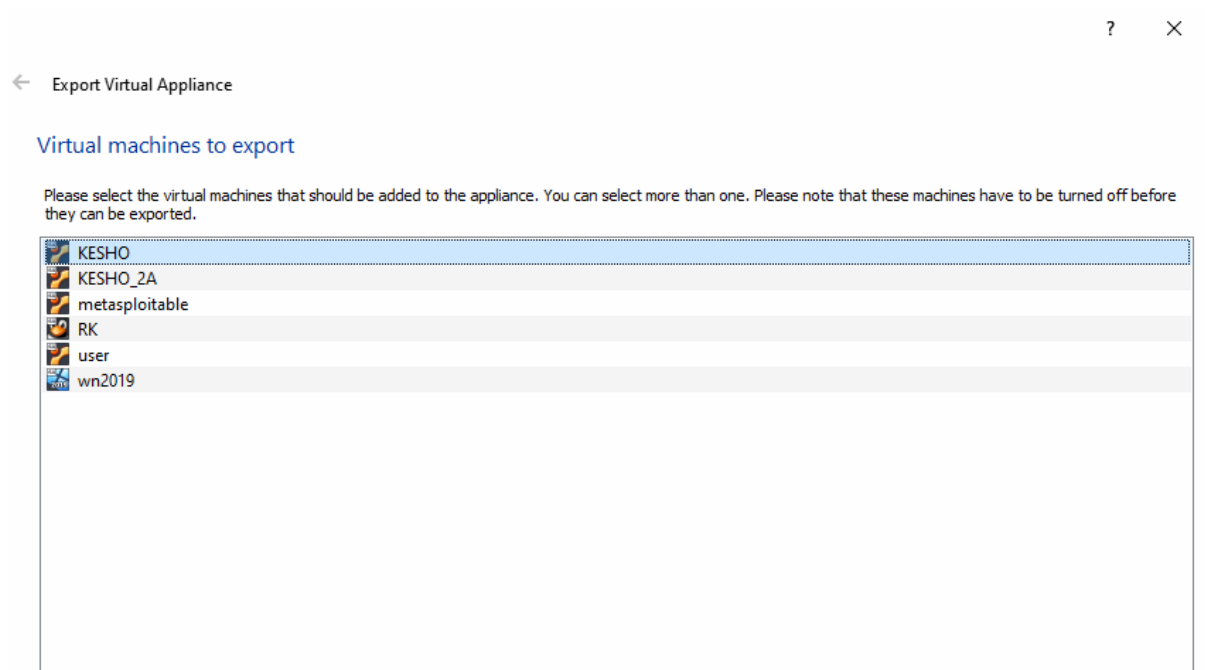


```
Activities  Terminal  Sat 14:03
keshoo@tce-VirtualBox: ~
File Edit View Search Terminal Help
keshoo@tce-VirtualBox:~$ cat kesho.java
public class kesho{
    public static void main(String args[]){
        long st=System.nanoTime();
        long sum=0;
        for(long i=0;i<1000000000L;i++){
            sum+=i;
        }
        long et=System.nanoTime();
        double ds=(et-st)/1000000000.0;
        System.out.println("sum is "+sum);
        System.out.println("Execution Time "+ds+" seconds");
    }
}
keshoo@tce-VirtualBox:~$
keshoo@tce-VirtualBox:~$
```

2) Shut down the VM completely. Open VirtualBox and select the VM Click File > Export Appliance (or File > Export OVF).



3) Select the vm which you want to export



4) Select the format which you want to export ova or ovf file format and select the path where you want to export in the file system

← Export Virtual Appliance

Appliance settings

Please choose a format to export the virtual appliance to.

The **Open Virtualization Format** supports only **ovf** or **ova** extensions. If you use the **ovf** extension, several files will be written separately. If you use the **ova** extension, all the files will be combined into one Open Virtualization Format archive.

The **Oracle Cloud Infrastructure** format supports exporting to remote cloud servers only. Main virtual disk of each selected machine will be uploaded to remote server.

Format: Open Virtualization Format 1.0

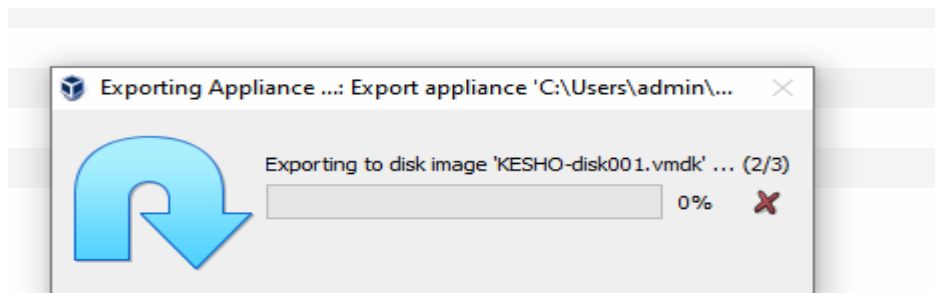
Please choose a filename to export the virtual appliance to. Besides that you can specify a certain amount of options which affects the size and content of resulting archive.

File: C:\Users\admin\Documents\KESHO.ova

MAC Address Policy: Include only NAT network adapter MAC addresses

Additionally: ☒ Write Manifest file
☐ Include ISO image files

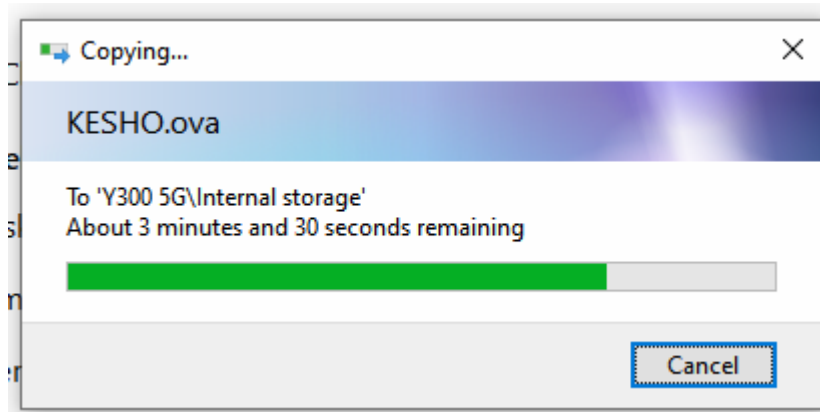
5) Click export to (export appliance to the file system where you selected the location)



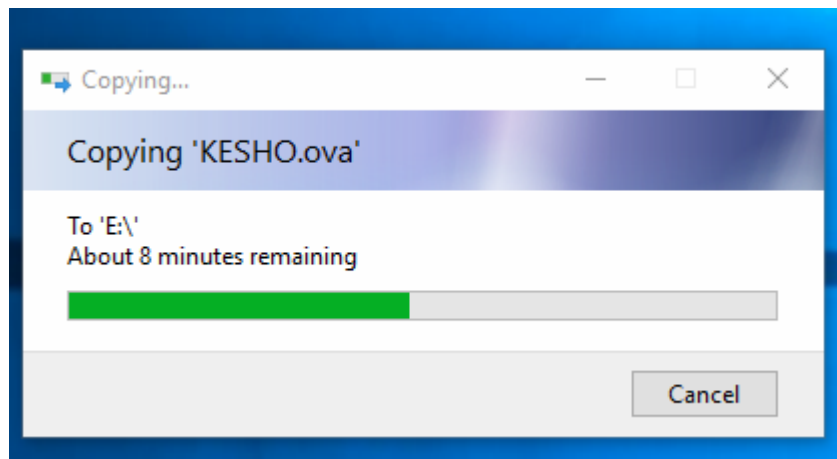
6) Check the folder whether the ova image has been present in the folder

Name	Date modified	Type	Size
Custom Office Templates	15-04-2025 16:33	File folder	
Amir.ova	16-07-2025 15:18	OVA File	9 KB
Computing Paradigms.docx	04-07-2025 12:41	Microsoft Word D...	199 KB
Default.rdp	19-07-2025 09:22	Remote Desktop ...	3 KB
KESHO.ova	19-07-2025 14:30	OVA File	78,30,788 KB

7) Copy the image to the external device via internet , USB etc I am using the usb and stored in my smartphone y3005G.



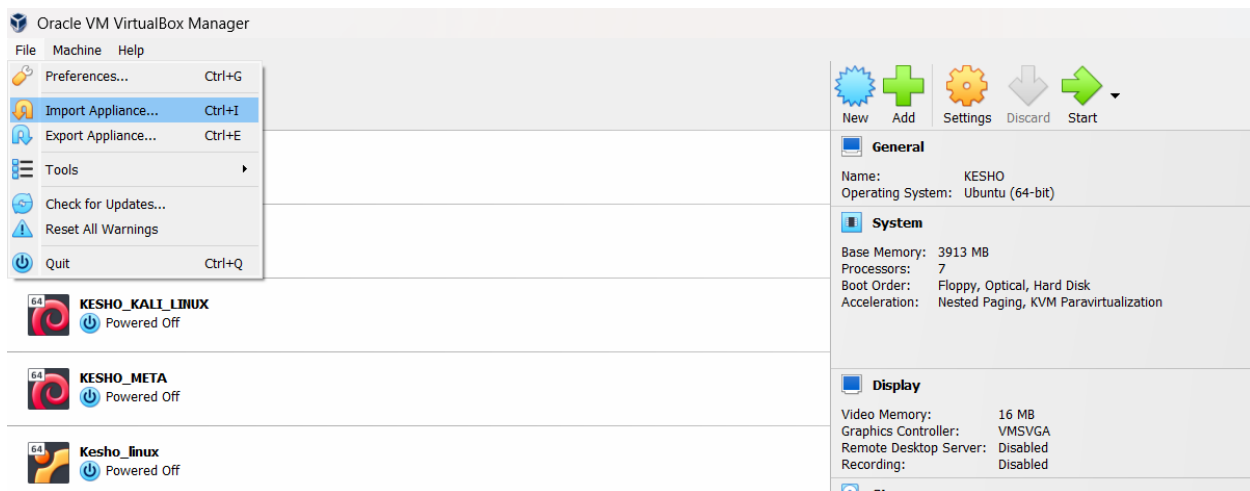
8) Copy the image from your external device to the destination machine via usb



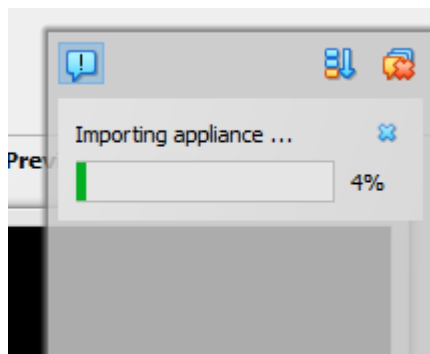
8) On the Destination Host: Import the OVA Open VirtualBox Go to File > Import Appliance (or File > Open). Browse and select the .ova file you transferred.

New folder	5/21/2025 12:27 PM	File folder	
padma-laptop	5/21/2025 1:32 PM	File folder	
project	8/1/2024 2:50 PM	File folder	
thangam	10/7/2024 11:53 AM	File folder	
MS OFFICE 2016 PRO.rar	4/2/2025 3:09 PM	WinRAR archive	2,481,266 KB
perumal .ova	7/18/2025 5:42 PM	OVA File	6,905,668 KB
KESHO.ova	7/19/2025 2:30 PM	OVA File	7,830,788 KB

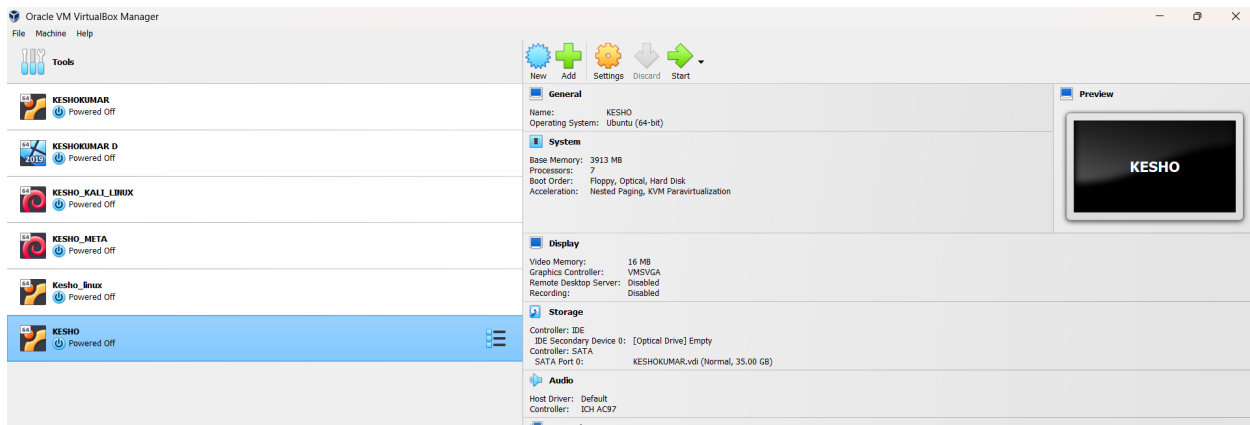
9) Import the image by import appliance in the file ->import appliance



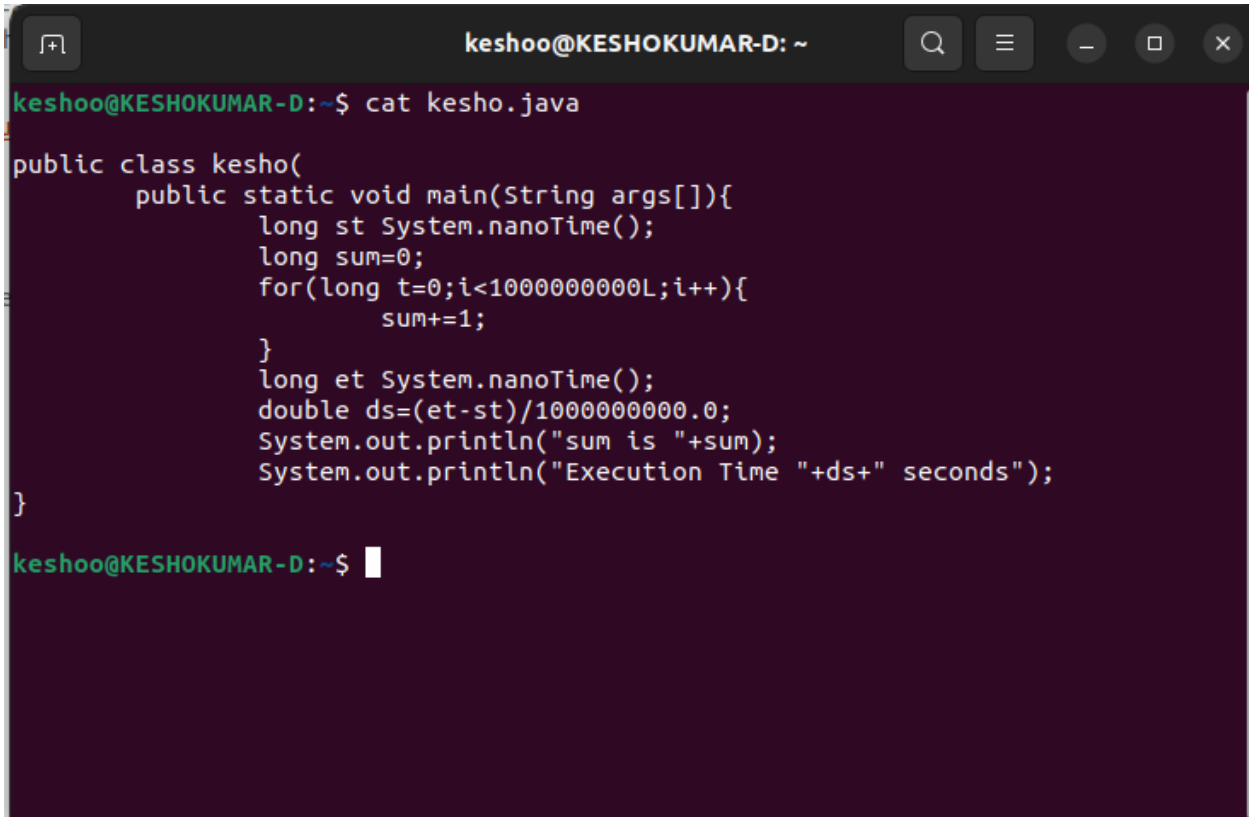
10)Now the image has been importing



11)Check whether the vm is shown in the oracle virtual box and open it



12)Open the java file which you saved first using the linux commands cat kesho.java
And its verified that the file are located and cold migration has done

A terminal window titled 'keshoo@KESHOKUMAR-D: ~' with standard Linux window controls. The command 'cat kesho.java' has been executed, displaying the contents of the file. The code is a Java class named 'kesho' with a 'main' method. It uses 'System.nanoTime()' to measure execution time and 'System.out.println()' to output the sum and execution time. The sum is calculated by iterating from 0 to 10,000,000,000L.

```
keshoo@KESHOKUMAR-D:~$ cat kesho.java
public class kesho(
    public static void main(String args[]){
        long st System.nanoTime();
        long sum=0;
        for(long t=0;i<10000000000L;i++){
            sum+=1;
        }
        long et System.nanoTime();
        double ds=(et-st)/1000000000.0;
        System.out.println("sum is "+sum);
        System.out.println("Execution Time "+ds+" seconds");
    }
keshoo@KESHOKUMAR-D:~$
```

RESULT:

The virtual machine was successfully migrated to the destination host in a powered-off state(cold migration) using the OVA export/import method. All configurations and data were retained, and the VM booted up and functioned correctly on the new host.